

What is the Relationship between Pollution and Population Growth Rate?

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INTRO

For our final project, we decided to focus on environmental issues. Our group's main goal was to explore the question: Is there a relationship between pollution and the human population growth rate in the United States, China and India?

We understand that by nature this question is hard to interpret in a causal manner, but we were curious to see if we could learn anything about the correlation by analyzing data of country population growth and pollution. For our analysis, we selected a sample of three countries- the United States, China, and India- because of their large populations and high levels of pollution in recent years.

To explore this relationship, we analyzed each country's annual population growth rate with several forms of pollution including waste incinerated, littered and mismanaged plastic waste, landfill waste, and total polluting waste. By comparing these variables across countries and over time, our goal was to identify the patterns or relationships that help to answer how environmental pollution and population growth rate in these three countries are correlated.

Our Hypothesis: If the amount of pollution increases over time, then the population growth rate in the United States, China, and India will also increase.

Our Result: For every 1% increase in the percentage of waste becoming a pollutant, India's population growth rate increases by about 0.1%. For every 1% increase in pollution, the U.S. population growth rate increases by about 0.23%. For every 1% increase in pollution, China's population growth rate increases by only 0.022%. This shows a positive correlation between population growth rate and pollution in these countries.

ANALYSIS

To analyze the relationship between pollution and population growth rates in the United States, China, and India, we first needed to gather data for both variables within the same time frame.

For pollution data, we used a dataset from Kaggle titled "3-share-plastic-fate.csv," created by Mohanadreza Momeni. This dataset includes seven columns: Entity, Code, Year, Share of waste recycled from total regional waste, Share of waste incinerated from total regional waste, Share of littered and mismanaged waste from total regional waste, and Share of waste landfilled from total regional waste. The dataset provides annual data from 2000 to 2019, which gives us a 20-year range to analyze.

Next, we collected population growth rate data for the United States, China, and India from the Federal Reserve Economic Data (FRED) website. FRED provided us with the annual population growth rates needed for our analysis.

Each dataset for the three countries contains two columns: "observation_date" and a country-specific population growth rate variable. These variables are: SPPOPGROWUSA for the United States, SPPOPGROWIND for India, and SPPOPGROWCHN for China.

The “observation_date” column provides the year, while the corresponding country code column shows the population growth rate for that year. However, because the pollution data is limited to the years 2000 through 2019, we also restricted our population growth rate data to that same period to keep it consistent. We made sure that the data we downloaded matched the 2000-2019 range used in our pollution dataset so that we could perform a direct year-by-year comparison between pollution levels and population growth. By aligning both datasets over the same time range, we were able to conduct a more accurate comparison and explore whether any patterns or correlations exist between increasing pollution levels and changes in population growth rates in the three countries.

Analysis of the graphs:

We mainly analyzed the relationship between population growth rates and various types of waste pollution in the United States, China and India from 2000 to 2019, and discovered several key trends.

First, Figure 1 (Appendix A) shows that from 2000 to 2018, all three countries- China, the United States, and India- have increased the share of their waste that is incinerated. China experienced the most dramatic rise, from around 9% to nearly 24%, surpassing the United States in around 2011. The United States saw a slower, steadier increase from approximately 15% to just under 19%. India began at a very low rate and gradually increased to about 4.3% by 2018. The upward trends indicate a growing reliance on incineration, which can reduce landfill use but may pose environmental and health risks due to air pollution. In this figure, we can see that the population growth rate is decreasing while the pollution is increasing, indicating a negative correlation. However, the units of incinerated waste in all countries are relatively low compared to other wastes, so we will need to look at other wastes in order to make a conclusion.

In Figure 1, we can also see a sudden drop in the share of waste incinerated for China in around 2017. This led us to believe that there may have been some policy put in place to manage how much waste China is incinerating. We found that in mid-2016, there were protests in southern China against garbage incineration (Gaphel, *The Tibet Express*). While we can't directly claim that this is the cause for the dip in waste incineration, it is likely a factor.

Figure 2 (Appendix A) indicates that litter and mismanaged waste has decreased in all three countries, especially in China and India, while the decline in the proportion of littered and mismanaged waste is consistent with the decline in the population growth rate. China's share dropped sharply from about 61% in 2000 to roughly 27% by 2018, while India reduced its share from around 68% to about 51%. The United States started with a much lower share, about 8% and continued to decline slightly over the period. This trend suggests that China and India have made considerable progress in formalizing and improving their waste management systems. This might also indicate that the improvement of environmental practices is related to the slowdown of population growth, especially in developing countries. In this figure, we can see that there is a positive correlation between population growth rate and pollution.

In Figure 3 (Appendix A), the United States maintains a consistently high reliance on landfilling, with rates hovering just above 74% throughout the entire period from 2000 to 2018. In contrast, both India and China show a steady increase in landfilled waste, rising from around 25% to nearly 38%. While increased landfilling may indicate improved containment of waste compared to open dumping, it still reflects a heavy dependence on a method that poses long-term environmental sustainability issues, especially if alternatives like recycling or composting are underused. Despite these differences, the population growth rates of these three countries are all declining. This indicates that the garbage disposal method itself may not have a significant impact on the growth trend. Meanwhile, the increasing reliance

of developing countries on landfill sites may reflect broader environmental or policy changes that are affecting population dynamics.

In Figure 4 (Appendix A), it reveals a consistent decline in the share of polluting waste and the population growth rate in all three countries. India and China both reduced their total waste by approximately 8 percentage points. India from around 95% to 87%, and China from roughly 95.5% to about 88%. The United States saw a more modest decline from about 97.5% to just above 96%. The consistent downward trends across all countries suggest improved waste treatment and management practices, contributing to reduced environmental pollution over time. Among all the indicators, this data most strongly supports our hypothesis that a decrease in pollution levels may be related to a decrease in population growth.

In Figure 5, (Appendix A), we calculated the regression statistics for the total percentage of pollutants versus population growth rate. In the table under the graph of each country, we see that the slope is approximately 0.10 for India, 0.23 for the United States and 0.02 for China. These numbers indicate that there's a positive correlation between population growth rate and pollution.

CONCLUSION

After analyzing this data, we learned that while we cannot directly claim there is a causal relationship, we can generally conclude that as population growth rates decline in India, China, and the United States, the pollution from these countries will also decrease, which means that they are positively correlated. We also see that in India and China, the decline in littered and mismanaged waste is offset by a substantial increase in landfilling and incineration. We assume one factor of these changes is likely to be policy changes, especially as people protest specific kinds of pollution. Overall, we can say that while declining population growth may improve how different countries manage waste, it does not necessarily lead to a decrease in pollution.

These conclusions point to several possible directions for future research. More research could include studying the long term effects of different types of pollution, how economic development affects the predominant waste disposal method in a country, and potential alternatives of waste disposal. Additionally, we would like to add more control variables, such as data from villages, islands, and other areas less affected by pollution, and compare them to urban regions. This approach may improve the accuracy of our project.

APPENDIX A: Figures and Tables

Figure 1: Share of Waste Incinerated vs. Population Growth Rate

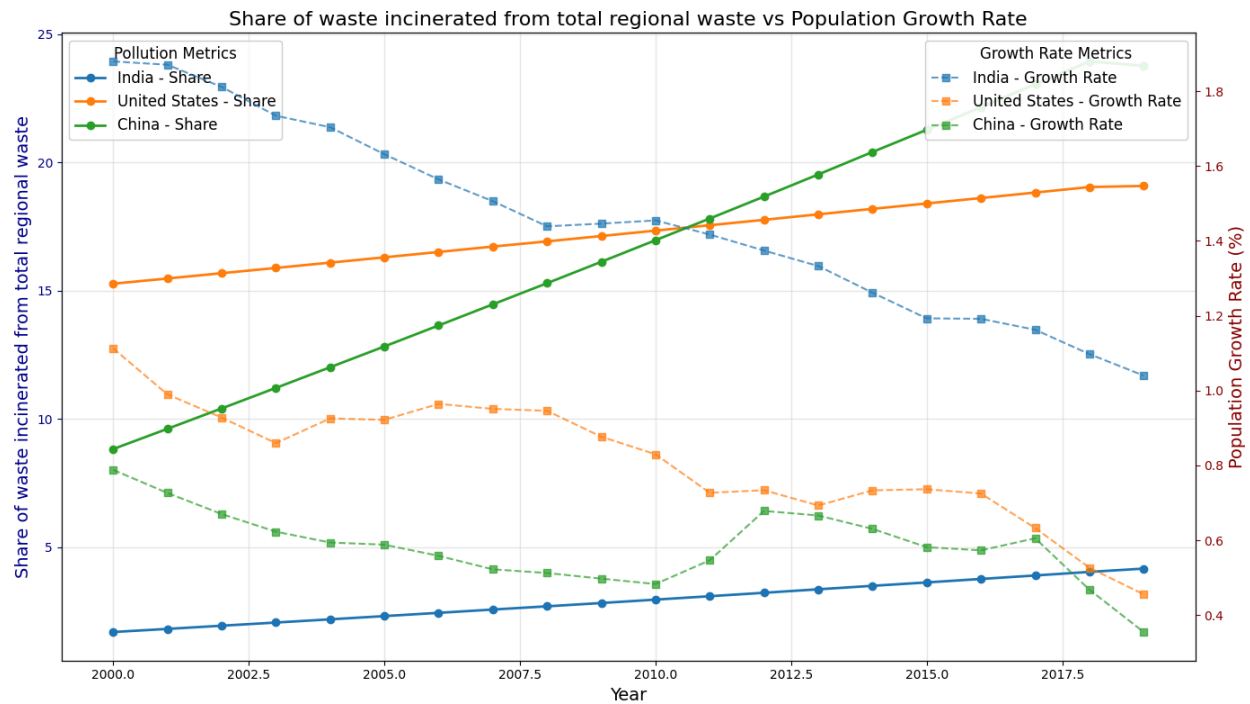


Figure 2: Share of Littered and Mismanaged Waste vs. Population Growth Rate

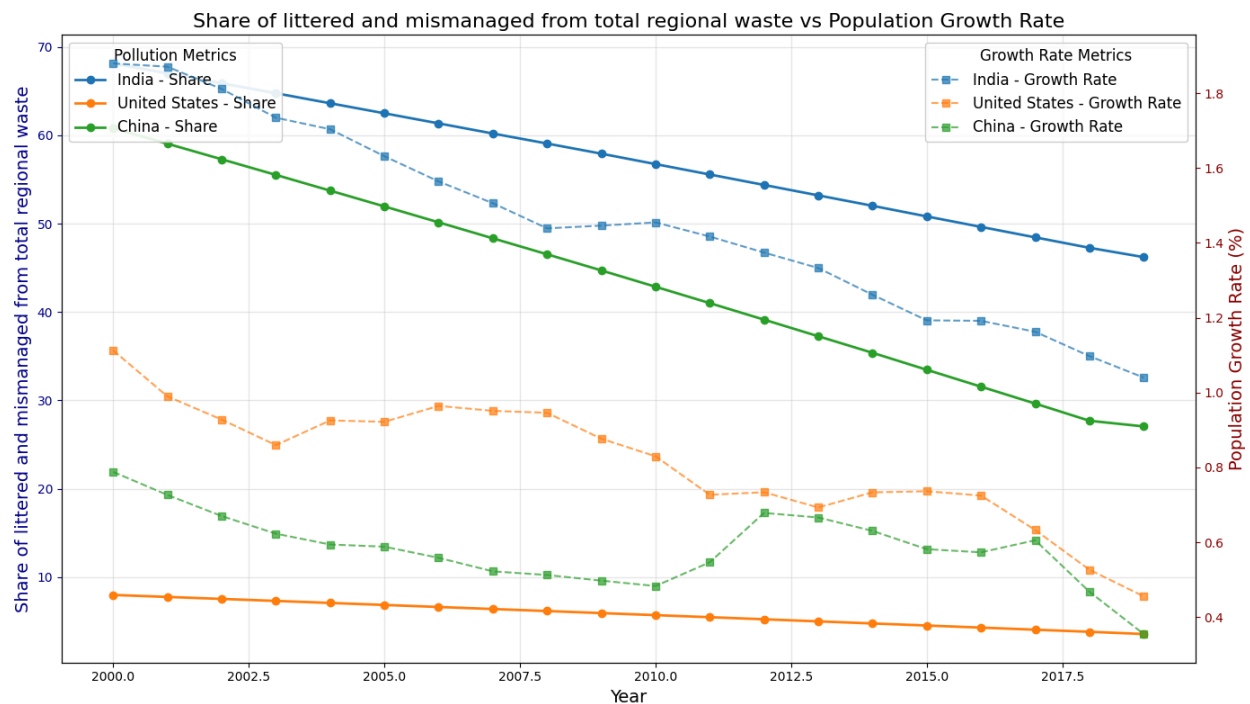


Figure 3: Share of Waste Landfilled vs. Population Growth Rate

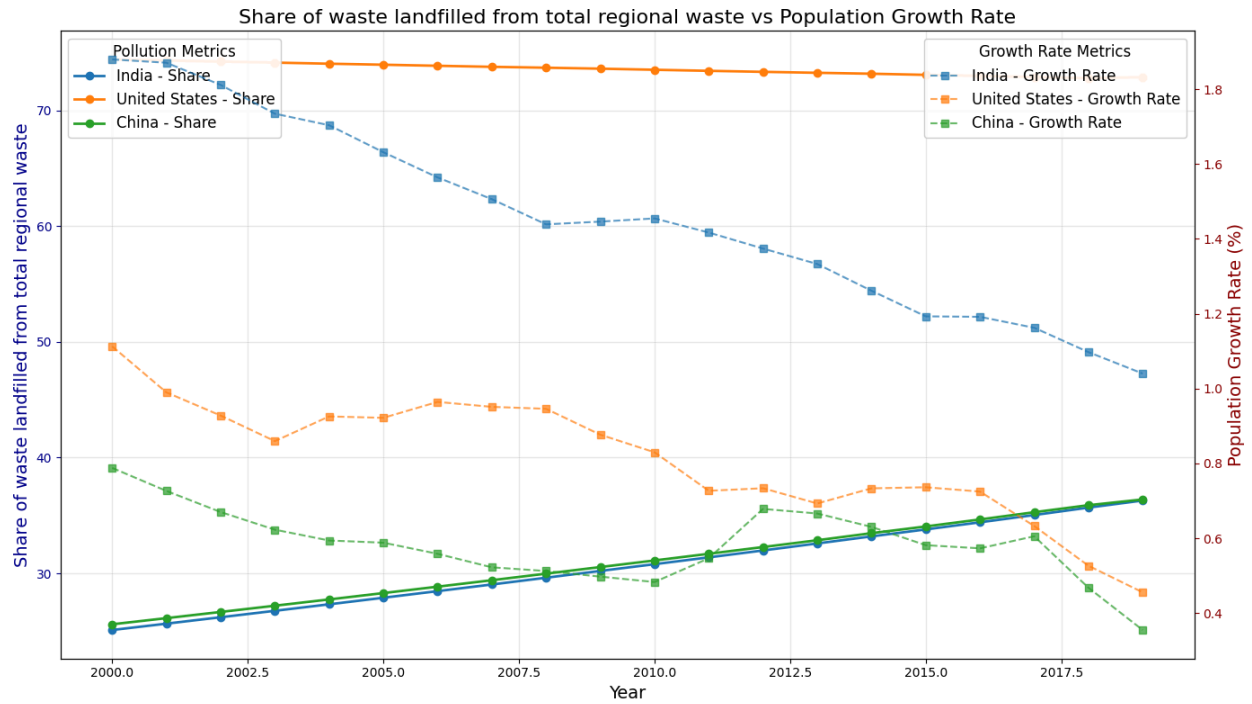


Figure 4: Total Pollutant Wastes vs. Population Growth Rate

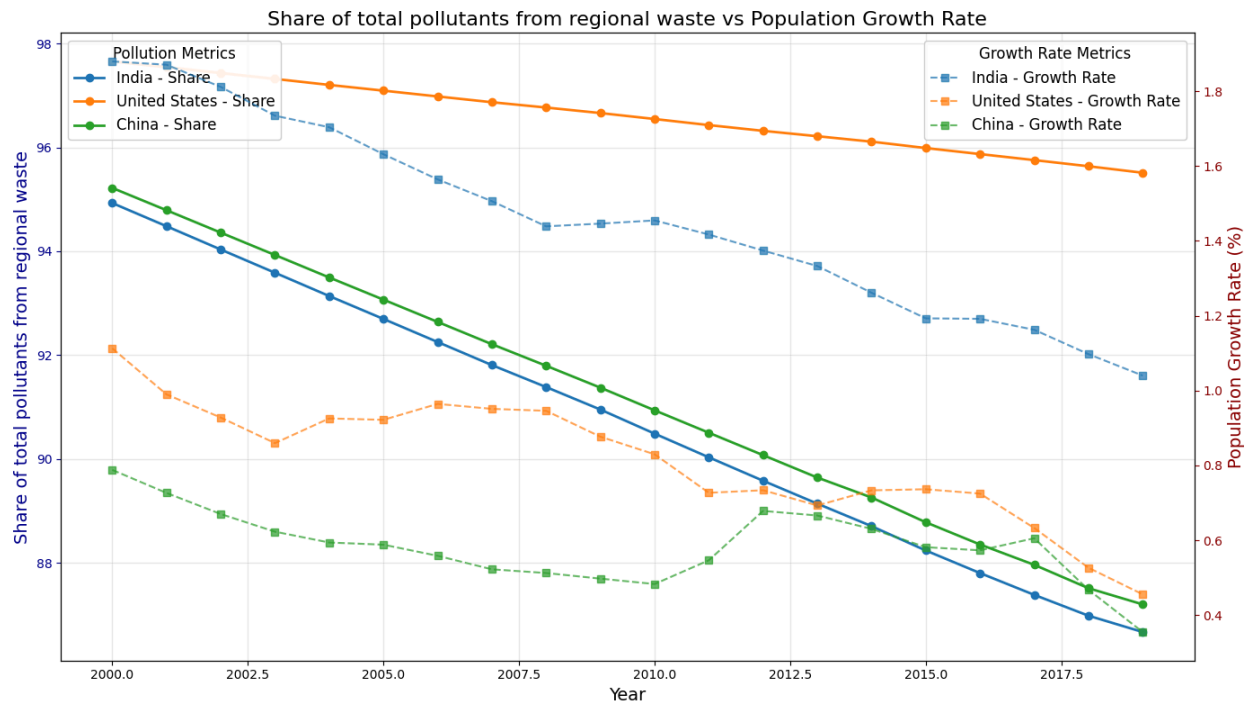
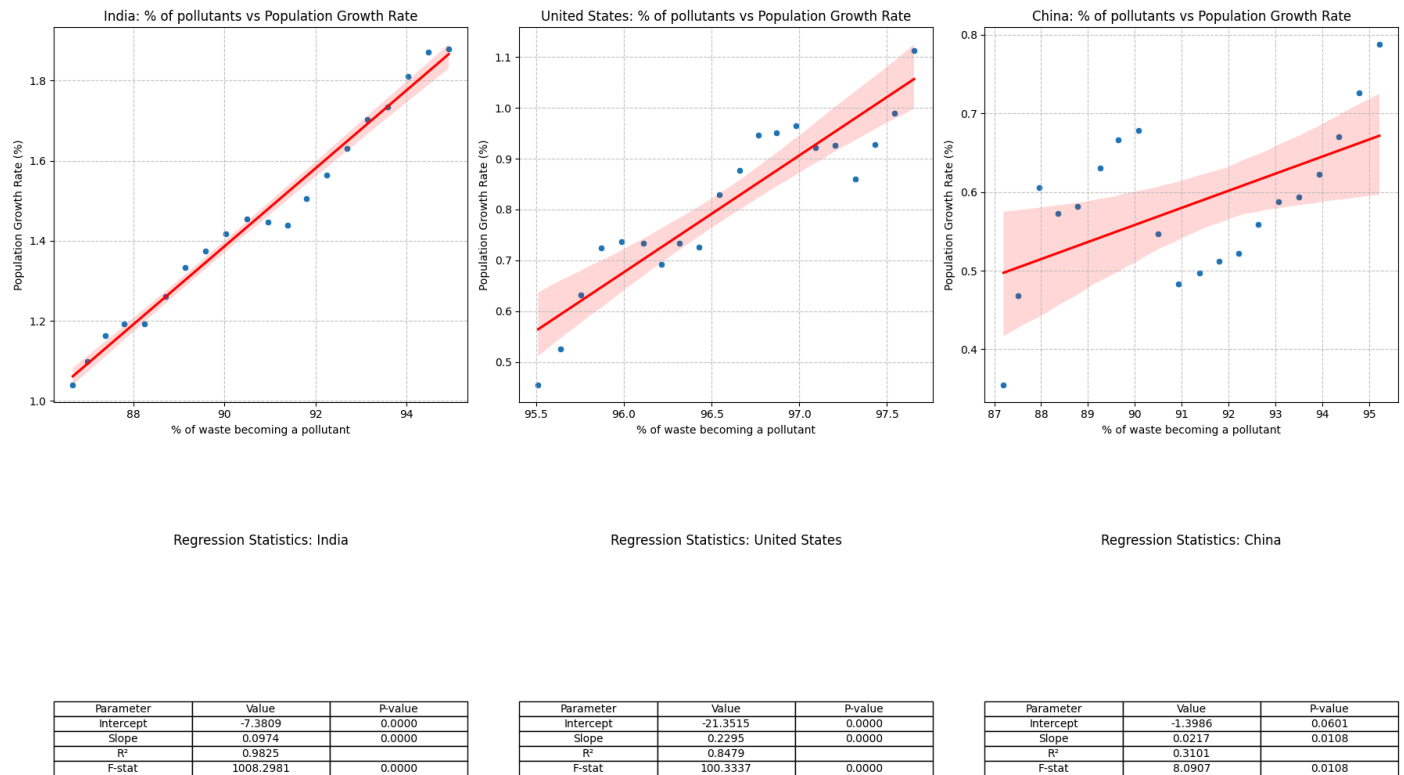


Figure 5: % of Pollutants vs. Population Growth Rate



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